

Tp4 report :

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Exercice 4 :

1- Identify the actors of the system :

Receptionnist , client , cashier, floor manager , Housekeeping staff , maintenance staff , Bank server.

2-Identify the actions that occur outside the system :

- The client makes a reservation.
- The client arrives at the hotel and, at the end of the stay, leaves it.
- The client consumes meals (breakfast, lunch, dinner) in the hotel restaurant.
- The client hands the final payment to the receptionist.
- The housekeeping staff clean the rooms.
- The floor manager inspects the rooms.
- The maintenance staff repair faults (broken air conditioner, water leak).
- The bank processes and authorizes the card payment on its own server.

3-Decomposition of the system into modules :

The system can be divided into four functional modules, each grouping the use cases that serve a common purpose.

The **Reservation management** module covers everything that happens before the client arrives. It allows the client to make a reservation by phone, through the web interface or through the mobile application, and it estimates the reservation amount. When this amount exceeds 5000 DH, the module also handles the online deposit payment, which is verified by the bank server. It also allows an existing reservation to be verified. The actors involved are the client and the bank server.

The **Stay management** module covers the client's presence in the hotel. It allows the receptionist to search for available rooms that match the client's preferences, to perform the check-in (which creates the stay record and changes the room status from "available" to "occupied"), and to perform the check-out at the end of the stay. The main actor is the receptionist.

The **Billing and services** module manages the financial side of the stay. It allows the receptionist to generate the invoice for the duration of the stay. It also allows the cashier to record the additional

consumptions (breakfast, lunch, dinner), whose charges are automatically added to the invoice. At check-out, the invoice is completed once the payment is received. The actors involved are the receptionist and the cashier.

The **Room management** module manages the state of the rooms after each stay. It allows the housekeeping staff to clean the rooms and update their status to “clean awaiting”. It then allows the floor manager to inspect each room and set it either to “available” or “unavailable” if a problem is found. Finally, it allows the maintenance staff to repair unavailable rooms so that they can be used again. The actors involved are the housekeeping staff, the floor manager and the maintenance staff.

4- The use-case diagram :

